

Laboratory Demo 3: Implementing a QPSK wireless link

EE161: Digital Communication Systems

San José State University

1 Introduction

The purpose of this experiment is to evaluate the performance of a wireless link implemented with ADALM-PLUTO radios and MATLAB Simulink models. Mapping is QPSK and pulses are square-root raised-cosine (SRRC). The effects that changes in the receiver gain and rolloff factor have on the spectral density and the bit-error rate (BER) are measured experimentally and analyzed.

You will need to download the following MATLAB Simulink models below from Canvas directory Files/Radio Experiments/Demo 3.

plutoradioQPSK-Tx_test_2020.slx

plutoradioQPSK-Rx_test_2020.slx

2 Link parameters

The default main parameters of the link are as follows:

- Software-Radio QPSK transmitter
 - SRRC filter rolloff factor, `alpha` = 1/2 (default)
 - Transmitter radio gain, `Gt` = -5 dB (default)
- Software-Radio QPSK receiver
 - SRRC filter rolloff factor, `alpha` = 1/2 (default)
 - Receiver radio gain, `Gr` = 25 dB (default)

For the report, find the values of length and number of samples per symbol of the SRRC filter as well the bit rate of the link.

3 Sensitivity to receiver gain

You must make sure that the center frequency `fc` is set up properly by typing `fc = VALUE`; in the command window, where `VALUE` is given by the instructor. This is needed to ensure that the wireless links do not use the same frequency range. **Use the same center frequency `fc` and frequency error values as in Demo 2.**

3.1 Receiver gain 25 dB (default value)

3.1.1 Transmitter

Open the QPSK transmitter radio model `plutoradioQPSK_Tx_test.slx` in one of the desktop computers (first student).

3.1.2 Receiver

Open the QPSK receiver model `plutoradioQPSK_Rx_test.slx` on the other desktop computer (second student). Wait for the transmitter model to run. Then start the receiver model and wait until it stops. This will take a couple of minutes in order to receive approximately 10 million bits.

Capture or sketch the received spectrum, the onstellation before and the constellation after correction. These appear in three figures. Attach the captures to your report. You may need to press the “scale” button in a window to see the full range of a plot. Take note also of the BER value (this is the top of the three numbers in the BER Display).

In the unlikely event that you get a high BER value (close to 0.5), try stop and restart the model. If that does not work, then try adding or subtracting in 500 Hz steps to the center frequency `fc`, in a similar manner as the channel sounding demo. As a last resort, increase the receiver gain by 5 dB at a time until you get a relatively low BER value.

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3.2 Receiver gain set to other values

Repeat the procedure in section 3.1 above with the receiver gain set as follows. Denote the receiver gain used in section 3.1 by `Grinit = 25` (or a higher value if this was needed). The receiver gain shall be set equal to each of the following values:

```
>> Gr = Grinit-20;  
>> Gr = Grinit-15;  
>> Gr = Grinit-5;  
>> Gr = Grinit+10;
```

For each receiver gain value, capture the spectral density of the received signal, the constellation plots, and BER.

4 Sensitivity to rolloff factor α

Repeat the procedure in section 3.1 with the receiver gain set back to 25 dB (or back to the value that gave a low BER) and the rolloff factor modified to each of the following values:

```
>> alpha = 0.1;  
>> alpha = 1;
```

For the report: What is the effect of the rolloff factor on the received spectrum? on the BER?

5 Performance analysis

The procedures in this section are to be done **after the experimental measurements are taken**. Attach the plots to your report.

5.1 Receiver gain vs. BER

Plot the receiver gain G_r (dB) versus BER in a semi-logarithmic graph.

5.2 SNR vs BER

For each of the receiver gain values in section 3, use the received spectral density (Spectrum Analyzer window) to estimate the SNR (dB) as the difference between the flat portion of the raised-cosine spectrum and the error floor. Then plot the estimated SNR (dB) versus BER in a semi-logarithmic graph. How does it compare to theory?

5.3 Receiver gain vs. constellation

Analyze the effect that the receiver gain has on the corrected constellation points at the receiver. (These are shown in the “After Carrier Synchronizer” window.)